Trey Chase

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PROFILE

Data scientist and applied statistician with experience in statistical modeling, data engineering, and analytics across sports, biomechanics, and performance data.

Technical Skills: Python (pandas, scikit-learn, XGBoost, PyMC), R (Shiny, brms/Stan, tidyverse), SQL
Data & Engineering: Snowflake, Airflow, Kafka, PySpark, FastAPI, Git
Analytics: Bayesian Modeling, Machine Learning, Survival Analysis, Data Visualization, Biomechanics, Player Health

PROFESSIONAL EXPERIENCE

DATA ANALYST | National Basketball Association | Remote | *Oct 2025 - May 2026*

- Designed and deployed the organization's first two compliance dashboards for a new league-wide assessment program to drive player outreach and revise testing schedules.
- Optimized four biomechanics visualization dashboards to identify trends across large, longitudinal datasets to identify player injury risk.
- Developed a Python-based flight phase detection algorithm to identify key phases of jumping movements from biomechanical sensor data, enabling managers to extract player-level max/min metrics for comparison and injury-risk assessment.

DATA ANALYST INTERN | National Basketball Association | New York, NY | *Jun 2025 - Aug 2025*

- Engineered and processed 10+ GB of high-frequency experimental datasets, integrating frame-by-frame physical metrics and scalar measurements across multiple trials.
- Developed the organization's first four biomechanics analytics dashboards, providing advanced insight into player load distribution on the lower extremity joints over time.
- Optimized internal testing and scheduling processes for the NBA season, improving compliance monitoring of NBA teams.
- Recognized with the Intern MVP Award for outstanding performance, out of 71 candidates.

DATA SCIENTIST | Duke University Baseball | Durham, NC | *Aug 2024 - May 2026*

- Implemented a predictive outcomes model from batted ball metrics to assess player performance.
- Developed the first internal biomechanics dashboard, leveraging season-long weight room data to track and quantify player improvement.
- Mentored undergraduates in developing skills in R programing, statistical theory, and applied analytics fundamentals.

DIRECTOR OF STATISTICS | San Luis Obispo Blues Baseball | San Luis Obispo, CA | *May 2024 - Aug 2024*

- Leveraged statistical methods, visualization, and internal software to manage organizational data.
- Informed strategic decision-making by analyzing Trackman data and streamlining scouting for the coaching staff.
- Helped lead the organization to its best result in three years, increasing its success rate by 19% year-over-year.

DATA ANALYST | Cal Poly San Luis Obispo Baseball | San Luis Obispo, CA | *Jan 2022 - Jun 2023*

- Constructed weekly analytical reports on external groups, utilizing large public datasets.
- Tracked and tagged performance data, captured and processed video and data submissions.
- Established a culture of data-informed decision-making within the organization.

PROJECTS

MASTER'S PORTFOLIO PROJECT | Duke University | *Jan 2026 - May 2026*

- Developed a hierarchical Bayesian linear model on a continuous outcome as a function of multiple mechanical and contextual covariates using brms/Stan on a multi-year dataset.
- Communicated baseball analytics findings to Duke coaching staff and presented technical statistical analyses to Duke Statistical Science faculty.
- Received Award for Best Portfolio in the Duke Statistical Science Program, out of 40 candidates.

MLB DAILY REPORTING DASHBOARD | Personal Project | *Jan 2026 - Present*

- Built and deployed an interactive dashboard summarizing game results, with automated reporting, leaderboards, and detail views showing percentile rankings and profiles for players.
- Engineered an automated daily pipeline integrating public APIs, precomputing datasets to cloud storage.
- Implemented AI agents to validate daily data refreshes and model results.

BASEBALL BIOMECHANICS MODELING | Driveline Baseball | *Aug 2024 - Dec 2024*

- Processed marker-based motion capture data for hitters and pitchers in Python using Driveline Baseball data.
- Fitted a Bayesian LASSO model to predict exit velocity for hitters and release speed for pitchers to assess player performance.
- Constructed a c3d animator to validate model results and visualize player mechanics for stakeholders.

EDUCATION

MASTER OF SCIENCE IN STATISTICAL SCIENCE | Duke University | *Aug 2024 - May 2026*
GPA: 3.70
Relevant Coursework: Bayesian Modeling, Time Series Forecasting, Applied Machine learning, Statistical Programming, Data Engineering, Causal Inference, Hierarchical Modeling

BACHELOR OF SCIENCE IN MATHEMATICS | California Polytechnic University San Luis Obispo | *Sep 2019 - Jun 2023*
GPA: 3.60
Relevant Coursework: Linear Algebra I-III, Differential Equations I-II, Calculus II-IV, Real Analysis, Complex Analysis, Microeconomics, Macroeconomics

AWARDS AND PROFESSIONAL MEMBERSHIPS

AMERICAN STATISTICAL ASSOCIATION
DUKE SPORTS ANALYTICS CLUB
MASTER'S PORTFOLIO | Awarded by Duke University Statistical Science Thesis Board
INTERN MVP | Awarded by NBA Intern Peers
DEAN'S LIST (x3) | Duke University
DEAN'S LIST (x5) | Cal Poly San Luis Obispo